

Heteroskedasticity and Clustered Standard Errors

Lecture 5

Recall that in a regression model,

$$y = x'\beta + u,$$

we assumed that $E(u|x) = 0$ and $V(u|x) = \sigma^2$. Heteroskedasticity occurs when $V(u_i|x) = \sigma_i^2$.

INSERT FIGURE HOMOCEDASTICITY-HETEROCEDASTICITY

If we use an estimator for $V(\hat{\beta})$ in the homocedastic case, we get a biased estimate of the true variance. Then, the classical inferential including the t and F tests, are invalidated since they use estimates of the variance of the estimator.

1 Generalized Least Squares

Suppose that $\mathbb{E}uu' = \Omega \neq \sigma^2 I_N$, and Ω is known. Consider a decomposition of the matrix $\Omega^{-1} = CC' = P\Lambda^{-1}P'$ where P is an orthogonal matrix and Λ is a diagonal matrix including the characteristic roots of Ω . It turns out that we can write

$$\mathbb{E}uu' = \Omega = (C')^{-1}C^{-1}.$$

We can then transform the regression as

$$\begin{aligned} C'y &= C'X\beta + C'u \\ \tilde{y} &= \tilde{X}\beta + \tilde{u} \end{aligned}$$

The error term vector $\tilde{u}$ is mean zero, and variance I (e.g. $\tilde{u} \sim \mathcal{N}(0, I)$) because the transformation achieves

$$V(\tilde{u}) = V(C'u) = C'V(u)C = C'(C')^{-1}C^{-1}C = I$$

Therefore, the generalized least squares estimator is

$$\hat{\beta} = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'\tilde{y} = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}y,$$

or equivalently,

$$\hat{\beta} = \arg \min ||y - X\beta||_{\Omega^{-1}}^2$$

The GLS estimator is unbiased with conditional variance equal to,

$$\begin{aligned} V(\hat{\beta}) &= (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}V(y|X)\Omega^{-1}X(X'\Omega^{-1}X)^{-1} \\ &= (X'\Omega^{-1}X)^{-1}, \end{aligned}$$

which is “smaller” than the variance-covariance matrix of the OLS estimator in a model of non-spherical errors.

Theorem 1. $V(\hat{\beta}_{OLS}) - V(\hat{\beta}_{GLS})$ is a positive semi-definite matrix.

Proof: For simplicity C is symmetric. Note first that the conditional variance of the OLS is equal to

$$V(\hat{\beta}_{OLS}) = V((X'X)^{-1}X'u) = (X'X)^{-1}X'\Omega X(X'X)^{-1}$$

Consider the decomposition,

$$\begin{aligned} X'\Omega^{-1}X &= (X'CC'X) = W'W \\ X'\Omega X &= (X'(C')^{-1}C^{-1}X) = S'S \\ X'X &= (X'CC^{-1}X) = W'S \end{aligned}$$

Then, we have that

$$(X'X)(X'\Omega X)^{-1}(X'X) = W'S(S'S)^{-1}S'W = W'P_S W \leq W'W$$

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The GLS estimator is the weighted least squares (WLS) estimator when we only have heterocedasticity,

$$C^{-1} = \Omega^{1/2} = \text{diag}(\sigma_i)$$

Therefore we can re-weight by $\Omega^{-1/2}$ instead of C .

1.1 Estimation

The GLS estimator is unfeasible meaning that it can not be implemented in practice. The matrix Ω is typically unknown, thus we would like to implement a method to estimate it. We

can consider instead $\Omega(\gamma)$, and after establishing a relation between the conditional variance and the parameter γ , we may be able to implement the method.

Consider for simplicity

$$y_i = x_i\beta + (z_i'\gamma)^{1/2}u_i,$$

which implies that the i th diagonal element of Ω is $\Omega_{ii} = (z_i'\gamma)$. Note that if we estimate γ , we can estimate Ω . A simple estimation procedure considers:

1. Obtain the OLS estimator of γ by running,

$$(y_i - x_i\hat{\beta})^2 = z_i'\gamma + v_i$$

2. Plug $\hat{\gamma}$ on Ω , and obtain

$$\hat{\beta} = (X'\hat{\Omega}^{-1}X)^{-1}X'\hat{\Omega}^{-1}y.$$

Other alternative convenient first steps exists in the literature for the multiplicative, quadratic, etc. models of heterocedasticity. For instance, the conditional variance of u could be assumed to be $\exp(\gamma'x)$.

For completeness, we review the consequences of ignoring heterocedasticity:

- The OLS estimator is unbiased but not efficient.
- The estimate of the variance of the OLS estimator is a biased estimator of the true variance. The classical testing procedures are invalidated.

2 Weighted Least Squares: A simple example

Consider the simple regression model without intercept,

$$y_i = \beta x_i + u_i \tag{1}$$

where the conditional variance is

$$V(u_i|x_i) = \sigma_i^2.$$

For simplicity we assume that

$$V(u_i|x_i) = \sigma_i^2 = \sigma^2 x_i$$

We use the simple model presented before to illustrate the effects of heteroscedasticity. The OLS estimator of β is,

$$\hat{\beta} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$$

The variance of the OLS estimator is in this case,

$$V(\hat{\beta}) = \frac{\sum_i^n x_i^2 \sigma_i^2}{(\sum_{i=1}^n x_i^2)^2}$$

We now consider an estimator that weighs each observation by $1/\sqrt{x_i}$,

$$\frac{y_i}{\sqrt{x_i}} = \beta \frac{x_i}{\sqrt{x_i}} + \frac{u_i}{\sqrt{x_i}} \Rightarrow y_i^* = \beta x_i^* + u_i^* \quad (2)$$

because $V(u_i/\sqrt{x_i}|x_i) = x_i^{-1}V(u_i|x_i) = x_i^{-1}\sigma^2 x_i = \sigma^2$. Consequently, the estimator of the slope parameter is,

$$\tilde{\beta} = \frac{\sum_{i=1}^n x_i^* y_i^*}{\sum_{i=1}^n x_i^{*2}}$$

and the variance of $\tilde{\beta}$ is,

$$V(\tilde{\beta}) = \frac{\sigma^2}{\sum_{i=1}^n x_i^{*2}} = \frac{\sigma^2}{\sum_{i=1}^n (x_i/\sqrt{x_i})^2}$$

Now we can compare the variance of the two estimators to see if, under heteroscedasticity, the OLS estimator is BLUE (given that it is unbiased we only have to check if it is minimum variance). Taking the ratio of the two estimators,

$$\frac{\text{Var}(\hat{\beta})}{\text{Var}(\tilde{\beta})} = \frac{\sum_{i=1}^n x_i^3 \sum_{i=1}^n x_i}{(\sum_{i=1}^n x_i^2)^2},$$

Recall that the squared correlation coefficient between a and b is

$$r^2 = \frac{(\sum ab)^2}{(\sum a^2)(\sum b^2)} \leq 1 \Rightarrow (\sum ab)^2 \leq (\sum a^2)(\sum b^2)$$

The last expression is a version of the Cauchy-Schwarz Inequality. Note that in our case, $a^2 = x^3$ and $b^2 = x$. Then,

$$\frac{\text{Var}(\hat{\beta})}{\text{Var}(\tilde{\beta})} \geq 1,$$

which implies that the OLS estimator is inefficient under heteroscedasticity.

3 Testing Heteroscedasticity:

We start assuming a multivariate regression model,

$$y_i = x_i' \beta + u_i$$

We assume that the null hypothesis is

$$H_0 : V(u|x) = \sigma^2,$$

that is, our null hypothesis is no heteroscedasticity. Since $E(u|x) = 0$, we write H_0 as

$$H_0 : E(u^2|x) = E(u^2) = \sigma^2,$$

Note that we need to evaluate the conditional (on the independent variables) expectation of the squared of the error term,

$$u^2 = \alpha_0 + \alpha_1 x_1 + \dots + \alpha_p x_p + \varepsilon,$$

The homocedastic case implies that $\alpha_1 = \alpha_2 = \dots = \alpha_p = 0$, therefore

$$E(u^2|x_1, \dots, x_p) = \alpha_0$$

Note however this is unfeasible because we do not know the error term u^2 . A simple approach is to replace it by $\hat{u}^2$ and estimate,

$$\hat{u}_i^2 = \alpha_0 + \alpha_1 x_{1i} + \dots + \alpha_p x_{pi} + \varepsilon_i$$

and test if the slope coefficients are all equal to zero,

$$H_0 : \alpha_1 = \alpha_2 = \dots = \alpha_p = 0; \quad H_a : \text{At least one } \alpha_j \neq 0$$

using a F test. If $F > c$, where c is determined by a value of statistical significance, we reject the null hypothesis of no heterocedasticity.

The Lagrange multiplier (LM) version the test is very simple to compute since

$$LM = n \times R_{\hat{u}^2}^2,$$

which is distributed as χ_p^2 under the null hypothesis of homocedasticity. This is the Breusch-Pagan (1980) test for heteroscedasticity.

A variation of the test was developed by White (1980). To simplify the presentation, assume that $p = 2$:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i,$$

The White's test is based on the estimation of

$$u_i^2 = \alpha_0 + \alpha_1 x_{1i} + \alpha_2 x_{2i} + \alpha_3 x_{1i}^2 + \alpha_4 x_{2i}^2 + \alpha_5 x_{1i} x_{2i} + \varepsilon_i,$$

The null hypothesis in this case is,

$$H_0 : \alpha_1 = \alpha_2 = \dots = \alpha_5 = 0; \quad H_a : \text{at least one } \alpha_j \neq 0,$$

which can be evaluated with a *LM* or *F* test statistic.

4 A heteroscedasticity-robust matrix

The covariance matrix of the error term, under homocedasticity, is equal to,

$$\Sigma = E(uu') = E(uu'|x) = E(\text{diag}(u_1^2, u_2^2, \dots, u_n^2)|x)$$

or alternatively,

$$\Sigma = \begin{bmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_n^2 \end{bmatrix}$$

Therefore, under heteroskedasticity, the variance of the estimator is,

$$\begin{aligned} V(\hat{\beta}|X) &= V((X'X)^{-1}X'y) \\ &= (X'X)^{-1}X'V(y|X)X(X'X)^{-1} \\ &= (X'X)^{-1}X'V(u|X)X(X'X)^{-1} \\ &= (X'X)^{-1}X'\Sigma X(X'X)^{-1} \end{aligned}$$

If we know the error term, then notice that $(uu')_{ii} = u_i^2$ and the conditional expected value of the variance is,

$$\begin{aligned} E(V(\hat{\beta}|X)) &= (X'X)^{-1}X'E(\text{diag}(u_1^2, u_2^2, \dots, u_n^2|X)X(X'X)^{-1} \\ &= (X'X)^{-1}X'\Sigma X(X'X)^{-1} \end{aligned}$$

In practice, we obtain least square residuals, $\hat{u}_i^2 = (y_i - x_i'\hat{\beta})^2$, where $\hat{\beta}$ is the least square

estimator of β . We can then form a matrix,

$$\hat{\Sigma} = \begin{bmatrix} \hat{u}_1^2 & 0 & \dots & 0 \\ 0 & \hat{u}_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \hat{u}_n^2 \end{bmatrix},$$

and construct a feasible estimator:

$$\hat{V}(\hat{\beta}|X) = (X'X)^{-1}X'\hat{\Sigma}X(X'X)^{-1}.$$

This estimator is called ‘heteroscedasticity-robust’ estimator of the covariance matrix. It was introduced to the econometrics literature in an influential paper by White (1980). There are other alternatives typically implemented in statistical packages for $\hat{u}^2$ (e.g., Andrews 1991). To simplify this, if $p = 1$, White (1980) suggests to obtain the standard error of the OLS estimator

$$se(\hat{\beta}_1) = \left(\frac{\sum_{i=1}^n (x_i - \bar{x})^2 \hat{u}_i^2}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2} \right)^{1/2}.$$

Note that the estimator represents a consistent estimator of the unknown covariance. In small samples, unfortunately, the estimator might be biased. To see this, recall that $\hat{u}\hat{u}' = uMu' = yMy'$. Then,

$$\begin{aligned} \hat{u}_i^2 &= (y_i - x_i'\hat{\beta})^2 = (y_i + x_i'\beta - x_i'\beta - x_i'\hat{\beta})^2 \\ &= (u_i - x_i'(\hat{\beta} - \beta))^2 \\ &= (u_i - x_i'(X'X)^{-1}X'u)^2 \\ &= u_i^2 - 2u_ix_i'(X'X)^{-1}X'u + x_i'(X'X)^{-1}X'uu'X(X'X)^{-1}x_i \end{aligned}$$

Under homocedasticity, and denoting the so-called leverage point $h_{ii} = x_i'(X'X)^{-1}x_i$, we have that,

$$\begin{aligned} E(\hat{u}_i^2) &= E(u_i^2 - 2u_ix_i'(X'X)^{-1}X'u + x_i'(X'X)^{-1}X'uu'X(X'X)^{-1}x_i) \\ &= E(u_i^2) - 2E(u_ix_i'(X'X)^{-1}X'u) + E(x_i'(X'X)^{-1}X'uu'X(X'X)^{-1}x_i) \\ &= \sigma^2 - 2\sigma^2h_{ii} + \sigma^2h_{ii} \\ &= \sigma^2(1 - h_{ii}) < \sigma^2 = E(u^2) \end{aligned}$$

Then, insisting with the simple example when $p = 1$, we have that,

$$\begin{aligned} E(\hat{V}(\hat{\beta}_1)) &= \frac{\sum_{i=1}^n (x_i - \bar{x})^2 E(\hat{u}_i^2)}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2} \\ &= \frac{\sum_{i=1}^n (x_i - \bar{x})^2 \sigma^2 (1 - h_{ii})}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2} \\ &= \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} - \frac{\sum_{i=1}^n (x_i - \bar{x})^2 h_{ii} \sigma^2}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2} \end{aligned}$$

indicating that $\hat{V}(\hat{\beta}|X)$ is downward bias (note that the first term is the variance of the least squares estimator under homocedasticity). An standard correction is to divide $\hat{u}_i^2$ by $1 - h_{ii}$.

5 Consistent Estimation of S

Define,

$$\hat{S} := \frac{1}{n} \sum_{i=1}^n \hat{u}_i^2 x_i x_i',$$

where $\hat{u} = y - x_i' \hat{\beta}$, and $\hat{\beta}$ is a consistent estimator for the parameter β .

To show the consistency of $\hat{S}$ for $S = E(x_i x_i' u_i^2)$ we need a condition on the fourth moment of x_i as stated before in Assumption A5.

Theorem 2. (*Consistency of $\hat{S}$*) Suppose S exists and it is finite and $\hat{\beta}$ is a consistent estimator for β . Then, $\hat{S} \rightarrow S$.

The formal proof will be discussed later, but I would like to provide a brief illustration considering $p = 1$. By definition,

$$\begin{aligned} \hat{u}_i^2 &= (y - x_i \hat{\beta})^2 = (y - x_i \beta + x_i \beta - x_i \hat{\beta})^2 = (u_i - x_i(\hat{\beta} - \beta))^2 \\ &= u_i^2 - 2u_i x_i(\hat{\beta} - \beta) + (\hat{\beta} - \beta)^2 x_i^2, \end{aligned}$$

and therefore,

$$\hat{u}_i^2 - u_i^2 = -2u_i x_i(\hat{\beta} - \beta) + (\hat{\beta} - \beta)^2 x_i^2.$$

It follows that,

$$\begin{aligned}
 \hat{S}_n - S_n &= \frac{1}{n} \sum_{i=1}^n \hat{u}_i^2 x_i^2 - \frac{1}{n} \sum_{i=1}^n u_i^2 x_i^2 \\
 &= -2(\hat{\beta} - \beta) \frac{1}{n} \sum_{i=1}^n u_i^2 x_i^3 + (\hat{\beta} - \beta)^2 \frac{1}{n} \sum_{i=1}^n (x_i^2)(x_i^2) \\
 &= -2(\hat{\beta} - \beta) \frac{1}{n} \sum_{i=1}^n u_i^2 x_i^3 + (\hat{\beta} - \beta)^2 \frac{1}{n} \sum_{i=1}^n x_i^4.
 \end{aligned}$$

Therefore, under the existing assumptions,

$$\hat{S}_n = \frac{1}{n} \sum_{i=1}^n \hat{u}_i^2 x_i^2 \rightarrow E(u_i x_i^2) =: S,$$

because the moments are bounded and the estimator $\hat{\beta}$ is consistent for β under Assumptions A1-A5.

6 Clustered Standard Errors

This section presents a few ideas on the estimation of standard errors in the presence of “group” effects. Most of the corrections you’ll find below are available in statistical packages, but I thought it might be a good idea to have a lecture discussing specifically potential issues in empirical research.

Suppose we have

$$y_i = x_i' \beta + u_i \tag{3}$$

and we use OLS to estimate the unknown parameter β . It is known that, if $Var(u_i) = \sigma^2$,

$$\hat{\beta} = (X'X)^{-1} X'y \tag{4}$$

$$Var(\hat{\beta}) = \sigma^2 (X'X)^{-1}. \tag{5}$$

If $Var(u_i) = \sigma_i^2$, we have that,

$$Var(\hat{\beta}) = (X'X)^{-1} X' \Omega X (X'X)^{-1} \tag{6}$$

where $\Omega = E u u'$. In terms of estimation, it is known that the unknown parameters can be estimated by

$$\hat{\sigma}^2 = \left(\frac{1}{n} \sum_{i=1}^n \hat{u}_i^2 \right), \tag{7}$$

$$\hat{\Omega} = \text{diag}(\hat{u}_i^2), \quad (8)$$

where $\hat{u}_i = y_i - x_i' \hat{\beta}$ is a residual. Note that $\hat{\Omega} \rightarrow \Omega$ if $\hat{\beta} \rightarrow \beta$. Therefore, the robust standard errors relies in an asymptotic approximation and may be biased in small samples.

Suppose now that we have $\{(y_i, x_i)\}_{i=1}^n$, but we know that the model may vary by clusters $c = 1, \dots, C$:

$$y_{ic} = x_{ic}' \beta_c + u_{ic}. \quad (9)$$

Although it is reasonable to think that $\text{Cov}(u_{ic}, u_{is}) = 0$, it may be the case that

$$\text{Cov}(u_{ic}, u_{jc}) \neq 0. \quad (10)$$

We start assuming that the error term u_{ic} is not independent of x_{ic} . For instance, if $u_{ic} = \alpha_c + v_{ic}$, then $\text{Cov}(x_{ic}, \alpha_c) \neq 0$. In this case, clusters introduce biases if the model does not condition on α_c . Therefore, one might be interested in estimating,

$$E(y_{ic} | X_{ic}, \alpha_c) = x_{ic}' \beta + \alpha_c, \quad (11)$$

when the number of clusters are large. The model can be estimated by classical least-squares transformations.

In the case of random effects, we can consider stacking observations within clusters,

$$y_c = X_c \beta + u_c \quad (12)$$

where y_c is a vector of dimension $n_c \times 1$, and X_c is a $n_c \times p$ matrix. Let $n = \sum_c n_c$. Then,

$$\hat{\beta} = (X'X)^{-1} X'y = \left(\sum_{c=1}^C \sum_{i=1}^{n_c} x_{ic} x_{ic}' \right)^{-1} \sum_{c=1}^C \sum_{i=1}^{n_c} x_{ic} y_{ic}.$$

Notice that $\text{Var}(\hat{\beta}) \neq \sigma^2 (X'X)^{-1}$. The clustered-robust standard error is,

$$(X'X)^{-1} X' \hat{\Omega} X (X'X)^{-1} = \left(\sum_{c=1}^C X_c' X_c \right)^{-1} \left(\sum_{c=1}^C X_c' \hat{u}_c \hat{u}_c' X_c \right) \left(\sum_{c=1}^C X_c' X_c \right)^{-1} \quad (14)$$

where the residuals are $\hat{u}_c = y_c - X_c \hat{\beta}$. The problem with this correction is that we need $C \rightarrow \infty$ for consistency.

If we assume that $u_{ic} = \alpha_c + v_{ic}$, we can adopt the random effect framework by assuming that α and v are independent. Notice that

$$E(u_{ic} u_{jc}) = E(\alpha_c^2 + v_{ic} \alpha_c + v_{jc} \alpha_c + v_{ic} v_{jc}) \quad (15)$$

and then, if $i = j$,

$$E(u_{ic}u_{ij}) = \sigma_\alpha^2 + \sigma_v^2 = \sigma^2, \quad (16)$$

and, if $i \neq j$,

$$E(u_{ic}u_{jc}) = \sigma_\alpha^2 = \rho\sigma^2, \quad (17)$$

where $\rho = \sigma_\alpha^2/(\sigma_u^2 + \sigma_\alpha^2)$. By results similar to the Balesta - Nerlove estimator, we have that

$$\Sigma_c = E(u_c u_c') = \sigma^2((1 - \rho)I_c + \rho 1_c' 1_c) \quad (18)$$

It then follows that the variance of the $\hat{\beta}$ *OLS* is,

$$Var(\hat{\beta}) = \left(\sum_{c=1}^C X_c' X_c\right)^{-1} \sum_{c=1}^C \sigma^2 X_c' ((1 - \rho)I_c + \rho 1_c' 1_c) X_c \left(\sum_{c=1}^C X_c' X_c\right)^{-1}. \quad (19)$$

A particular useful case to fix ideas is a model with individual invariant variables. Suppose now that the model is,

$$y_{ig} = \beta_0 + \beta_1 x_g + u_{ig}, \quad (20)$$

It is possible to show that,

$$V(\hat{\beta}_1) = (1 + (M - 1)\rho)V(\tilde{\beta}_1) \quad (21)$$

where $V(\tilde{\beta}_1)$ is the variance of standard *OLS* estimator. Note that the bias of the *OLS* could be severe.

It is possible to test for cluster effects following Moulton (1997). The null hypothesis is

$$H_0 : \rho = \frac{\sigma_\alpha^2}{(\sigma_u^2 + \sigma_\alpha^2)} = 0 \quad (22)$$

$$H_a : \rho > 0. \quad (23)$$

A one-sided *LM* test can be formulated as follows:

$$LM = \frac{\sum_c (N_c \bar{u}_c)^2 - \sum_c \sum_i \hat{u}_{ic}^2}{\hat{\sigma}^2 [2(\sum_c N_c^2 - N)]^{1/2}} \quad (24)$$

where $\hat{\sigma}^2 = \sum_c \sum_i \hat{u}_{ic}^2 / N$, where $\hat{u}_{ic}$ denotes the least-square residual and $\bar{u}_c$ is the average residual for cluster C .

Example 1. Moulton (1986) wage regressions considering CPS data. The number of individuals in the sample was $n \approx 18,000$ in $c = 49$ states. Although the intraclass correlation coefficient estimate $\hat{\rho}$ was 0.03, the bias was large because the clusters were large

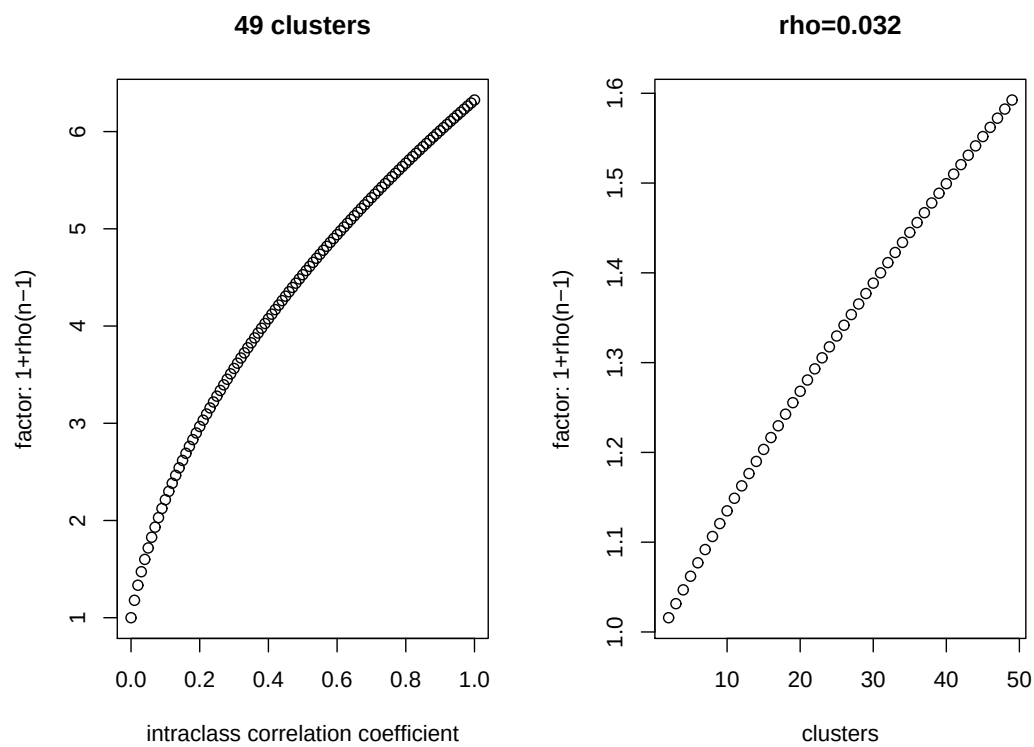


Figure 1: Standard Errors and the Moulton factor

($M \approx 380$). The Mounon factor, which is $((1 + (M - 1)\rho))^{.5}$, was 3.7 leading to sizable biases in the standard errors.